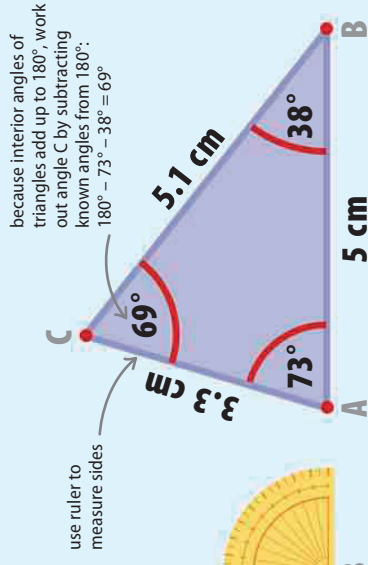
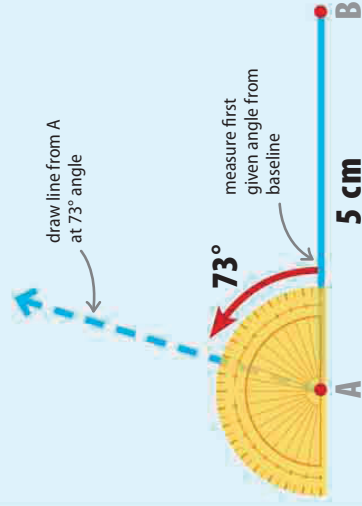


Constructing a triangle when two angles and one side are known (AAS)

A triangle can be constructed when the two angles, 73° and 38° , are given, along with the length of the side that falls between them, for example, 5 cm.



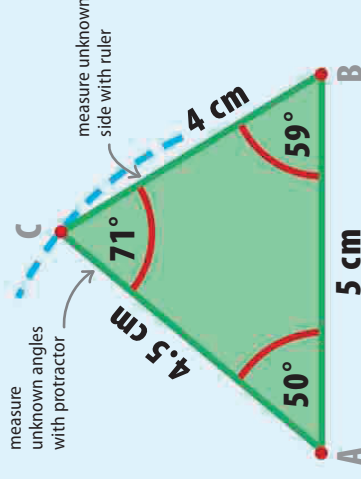
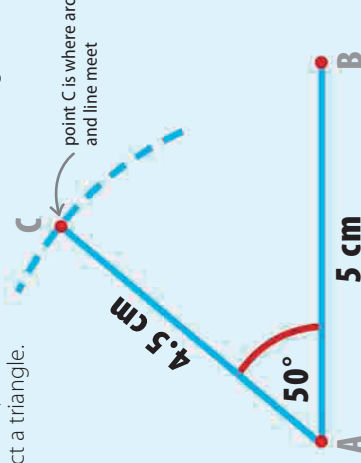
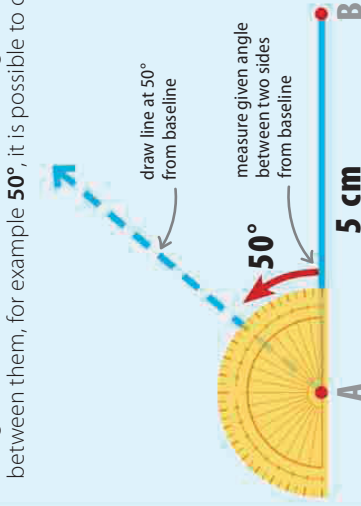
Draw the baseline of the triangle, here 5 cm. Label the ends A and B. Place the protractor over A and measure the first angle, 73° . Draw a side of the triangle from A.

Place the protractor over point B and mark 38° . Draw another side of the triangle from B. Point C is where the two new lines meet.

Join the points to complete the triangle. Calculate the unknown angle, and use a ruler to measure the two unknown sides.

Constructing a triangle when two sides and the angle in between are known (SAS)

Using the measurements for two of a triangle's sides, for example, 5 cm and 4.5 cm, and the angle between them, for example 50° , it is possible to construct a triangle.



Draw the baseline, using the longest length. Label the ends A and B. Place the protractor over point A and mark 50° . Draw a line from A that runs through 50° . This line will be the next side of the triangle.

Set the compass to the second length, 4.5 cm. Place the compass on point A and draw an arc. Point C is found when the arc intersects the line through point A.

Join the points to complete the triangle. Use a protractor to measure the unknown angles and a ruler to measure the length of the unknown side.